14-subgroup: 子群 (Subgroup)

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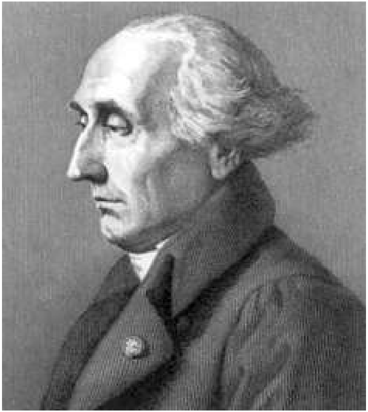
Lagrange's Theorem



Joseph-Louis Lagrange ($1736 \sim 1813$)

Fundamental Homomorphism Theorem

Emmy Noether ($1882 \sim 1935$)

Lagrange's Theorem

Help us understand the structure of a group
via its subgroups/normal subgroups

Fundamental Homomorphism Theorem

Definition (Coset (陪集))

Suppose that $H \leq G$. For $a \in G$,

$$aH = \{ah \mid h \in H\}, \quad Ha = \{ha \mid h \in H\},$$

is called the left coset (左陪集) and right coset of H in G, respectively.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \le S_3$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \le S_3$$

$$(1)H = H = (1\ 2)H \le S_3$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

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$$(1)A_3 = (1\ 2\ 3)A_3 = (1\ 3\ 2)A_3 = A_3 \leq S_3$$

$$(1\ 2)A_3 = (1\ 3)A_3 = (2\ 3)A_3 = \{(1\ 2), (1\ 3), (2\ 3)\}$$

Theorem

Suppose that $H \le G$, $a, b \in G$.

(1)
$$a \in aH$$

(2)
$$aH = H \iff a \in H$$

(3)
$$aH \le G \iff a \in H$$

(4)
$$aH = bH \iff a^{-1}b \in H$$

(5)
$$\forall a, b \in G\ (aH = bH) \lor (aH \cap bH = \emptyset)$$

(6)
$$|aH| = |H| = |bH|$$

$$\boxed{aH = H \iff a \in H}$$

$$\boxed{aH = H \iff a \in H}$$

$$\boxed{aH = H \implies a \in H}$$

$$\boxed{aH = H \iff a \in H}$$

$$\boxed{aH = H \implies a \in H}$$

$$a \in aH = H$$

$$\boxed{aH = H \iff a \in H}$$

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$$\boxed{aH = H \implies a \in H}$$

$$a \in aH = H$$

$$\boxed{a \in H \implies aH = H}$$

$$aH \subseteq HH \triangleq \{hh' \mid h, h' \in H\} = H$$

$$\boxed{aH = H \iff a \in H}$$

$$\boxed{aH = H \implies a \in H}$$

$$a \in aH = H$$

$$\boxed{a \in H \implies aH = H}$$

$$aH \subseteq HH \triangleq \{hh' \mid h, h' \in H\} = H$$

$$H = (aa^{-1})H = a(a^{-1}H) \subseteq aH$$

$$\boxed{aH \leq G \iff a \in H}$$

$$\boxed{aH \le G \iff a \in H}$$

$$\boxed{a \in H \implies aH \le G}$$

$$\boxed{aH \le G \iff a \in H}$$

$$\boxed{a \in H \implies aH \le G}$$

$$a \in H \implies aH = H \le G$$

$$\boxed{aH \le G \iff a \in H}$$

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$$\boxed{aH \le G \implies a \in H}$$

$$a \in aH \implies a^{-1} \in aH \implies e \in aH$$

$$\boxed{aH \le G \iff a \in H}$$

$$\boxed{a \in H \implies aH \le G}$$

$$a \in H \implies aH = H \le G$$

$$\boxed{aH \le G \implies a \in H}$$

$$a \in aH \implies a^{-1} \in aH \implies e \in aH$$

$$\exists h \in H\ e = ah \implies a = h^{-1} \in H$$

$$\boxed{aH = bH \iff a^{-1}b \in H}$$

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$$aH = bH \implies a^{-1}aH = a^{-1}bH \implies a^{-1}bH = H$$

$$a^{-1}bH = H \implies a(a^{-1}bH) = aH \implies bH = aH$$

$$\boxed{\forall a, b \in G \; (aH = bH) \vee (aH \cap bH = \emptyset)}$$

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$$\forall a, b \in G \; (aH \cap bH \neq \emptyset \to aH = bH)$$

$$\boxed{\forall a, b \in G \; (aH = bH) \lor (aH \cap bH = \emptyset)}$$

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Take any $g \in aH \cap bH$.

$$\boxed{\forall a, b \in G \; (aH = bH) \lor (aH \cap bH = \emptyset)}$$

$$\forall a, b \in G \; (aH \cap bH \neq \emptyset \to aH = bH)$$

Take any $g \in aH \cap bH$.

$$\exists h_1, h_2 \in H \; (ah_1 = g = bh_2)$$

$$\boxed{\forall a, b \in G\ (aH = bH) \lor (aH \cap bH = \emptyset)}$$

$$\forall a, b \in G\ (aH \cap bH \neq \emptyset \to aH = bH)$$

Take any $g \in aH \cap bH$.

$$\exists h_1, h_2 \in H\ (ah_1 = g = bh_2)$$

$$aH = a(h_1 H) = (ah_1)H = (bh_2)H = b(h_2 H) = bH$$

$$\boxed{|aH| = |H| = |bH|}$$

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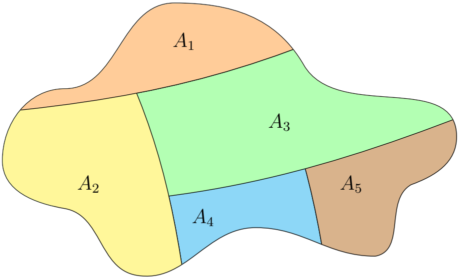
$$\sigma : aH \to H, \qquad ah \mapsto h$$

A balanced partition of G by its subgraph H



$$\boxed{\forall a, b \in G\ (aH = bH) \vee (aH \cap bH = \emptyset)}$$

$$\boxed{|aH| = |H| = |bH|}$$

Definition (Index (H 在 G 中的指标/指数))

$$G/H = \{gH \mid g \in G\}$$

$$[G : H] \triangleq |G/H|$$

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$$G/H = \{gH \mid g \in G\}$$

$$[G : H] \triangleq |G/H|$$

Theorem (Lagrange's Theorem)

Suppose that $H \leq G$*. Then*

$$|G| = [G : H] \cdot |H|$$

Theorem

$$H \leq G \implies |H| \mid |G|$$

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There are *no* subgraphs of order 5, 7, or 8 of a group of order 12.

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Theorem

Let G be a finite group, $a \in G$. Then the order of a divides the order of G.

Theorem

$$H \leq G \implies |H| \,|\, |G|$$

There are *no* subgraphs of order 5, 7, or 8 of a group of order 12.

Theorem

Let G be a finite group, $a \in G$. Then the order of a divides the order of G.

Theorem

- *There are only 2 groups of order 4.*
- *There are only 2 groups of order 6.*

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$$H(1) = H = H(1\ 2)$$
$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$
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It is possible that $aH \neq Ha$.

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$$(1)H = (1\ 2\ 3)H = (1\ 3\ 2)H = H \le S_3$$
$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

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$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

$$\forall a \in S_3\ aH = Ha$$

Definition (Normal Subgroup (正规子群))

Suppose that $H \leq G$. If

$$\forall a \in G \; aH = Ha,$$

then H is a normal subgroup of G, denoted $H \lhd G$.

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$$aH = Ha \not\Longrightarrow \forall h \in H \; ah = ha$$

$$aH = Ha \Longrightarrow \forall h \in H \; \exists h' \in H. \; ah = h'a$$

Theorem

$$H \lhd G \iff \forall a \in G, h \in H \; aha^{-1} \in H$$

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$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \; aha^{-1} \in H \end{aligned}$$

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$$a^{-1}ha = a^{-1}h(a^{-1})^{-1} \in H \implies ha \in aH \implies Ha \subseteq aH$$

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$$\forall \sigma \in S_3, \tau \in H.\ \sigma\tau\sigma^{-1} \in H$$

Theorem

$$\sigma\tau\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}$$

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$$(1\ 2)(1\ 2\ 3)(1\ 2)^{-1} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (1\ 3\ 2)$$

Definition (正规子群的陪集)

Suppose that $H \triangleleft G$.

$$G/H = \{aH \mid a \in G\}$$

is the set of cosets of H in G.

Definition (Quotient Group (商群))

Suppose that $H \triangleleft G$. Define

$$aH \cdot bH = (ab)H.$$

Then $(G/H, \cdot)$ is a group, called the quotient group of G by H (denoted G/H).

$aH \cdot bH = (ab)H$ is well-defined

$$aH \cdot bH = (ab)H \text{ is well-defined}$$

$$aH = a'H \land bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

结果与代表元的选取无关

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

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$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \lhd S_3$$

$$G/H = \{(1)H, (1\ 2)H\}$$

$$G = \mathbb{Z} \qquad H = 6\mathbb{Z} \lhd G$$

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$$G/H = \{0 + H, 1 + H, \ldots, 5 + H\}$$

Definition (Homomorphism (同态))

Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a function such that

$$\forall a, b \in G.\ \phi(ab) = \phi(a)\phi(b).$$

Then ϕ is a homomorphism from G to G'.

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Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a function such that

$$\forall a, b \in G.\ \phi(ab) = \phi(a)\phi(b).$$

Then ϕ is a homomorphism from G to G'.

If ϕ is a bijection, then G and G' are called isomorphic.

$$\phi : G \cong G'$$

$$\begin{aligned}\phi : \mathbb{Z} &\to \mathbb{R}^* \\ n &\mapsto (-1)^n\end{aligned}$$

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$$\phi(m+n) = (-1)^{m+n} = \phi(m)\phi(n)$$

$$\begin{aligned} \phi : \mathbb{Z} &\to \mathbb{Z}_6 \\ a &\mapsto [a]_6 \end{aligned}$$

$$\phi(a+b) = [a+b]_6 = \phi(a) + \phi(b)$$

$$\begin{aligned} \phi : \mathbb{R}[x] &\to \mathbb{R}[x] \\ f(x) &\mapsto f'(x) \end{aligned}$$

$\mathbb{R}[x]$：全体实系数多项式关于多项式的加法构成的群

$$\phi(f(x)+g(x)) = (f(x)+g(x))' = \phi(f(x)) + \phi(g(x))$$

Theorem

Suppose that ϕ is a homomorphism from G to G'.
Let e and e' are identities of G and G', respectively.

(1) $\phi(e) = e'$

(2) $\phi(a^{-1}) = (\phi(a))^{-1}$

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$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e) \implies \phi(e) = e'$$

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e' = \phi(a)(\phi(a))^{-1}$$

Theorem

Suppose that ϕ is a homomorphism from G to G'.

(1)

$$H \leq G \implies \phi(H) \leq G'$$

(2)

$$H \lhd G \implies \phi(H) \lhd \phi(G)$$

Theorem

Suppose that ϕ is a homomorphism from G to G'.

(1)

$$H \le G \implies \phi(H) \le G'$$

(2)

$$H \lhd G \implies \phi(H) \lhd \phi(G)$$

(3)

$$K \le G' \implies \phi^{-1}(K) \le G$$

(4)

$$K \lhd G' \implies \phi^{-1}(K) \lhd G$$

Definition (核 (Kernel))

Suppose that ϕ is a homomorphism from G to G'.
Let e' be the identity of G'.

$$\phi^{-1}(\{e'\}) = \{a \in G \mid \phi(a) = e'\}$$

is the kernel of ϕ, denoted $\operatorname{Ker} \phi$.

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$$\operatorname{Ker} \phi \lhd G$$

$$
\begin{aligned}
\phi : \mathbb{Z} &\to \mathbb{R}^* \\
n &\mapsto (-1)^n
\end{aligned}
$$

$$\mathrm{Ker}\ \phi =$$

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$$\text{Ker } \phi = 2\mathbb{Z}$$

$$
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$$\begin{aligned} \phi : \mathbb{R}[x] &\to \mathbb{R}[x] \\ f(x) &\mapsto f'(x) \end{aligned}$$

$\mathbb{R}[x]$：全体实系数多项式关于多项式的加法构成的群

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$$\text{Ker}\ \phi = \mathbb{R}$$

Theorem (Fundamental Homomorphism Theorem)

Suppose that ϕ is a homomorphism from G to G'. Then

$$G/Ker\,\phi \cong \phi(G).$$

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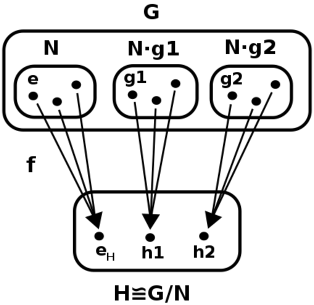
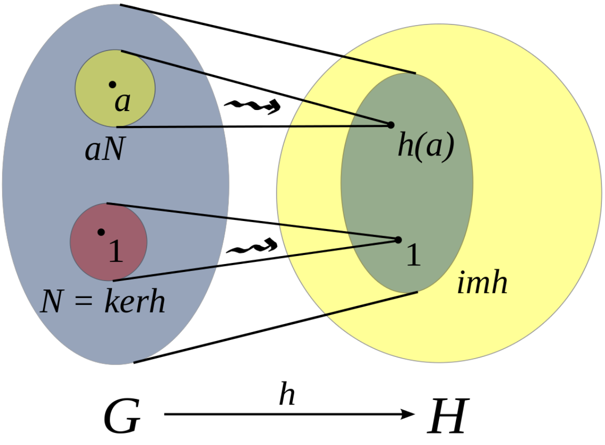


$$N = \text{Ker}\ \phi$$

$$G/(N \triangleq \mathrm{Ker}\ h) \cong (h(G) \triangleq \mathrm{im}\ h)$$



$$aN \mapsto h(a)$$

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\begin{aligned}
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$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\mathbb{Z}/(2\mathbb{Z}) = (2\mathbb{Z}, 2\mathbb{Z}+1) \cong \phi(\mathbb{Z}) = (-1, 1)$$

$$\phi : \mathbb{Z} \to \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

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$$\mathbb{Z}/(6\mathbb{Z}) = \{0 + H, 1 + H, \ldots, 5 + H\} \cong \phi(\mathbb{Z}) = \mathbb{Z}_6$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\to \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$：全体实系数多项式关于多项式的加法构成的群

$$\text{Ker}\ \phi = \mathbb{R}$$

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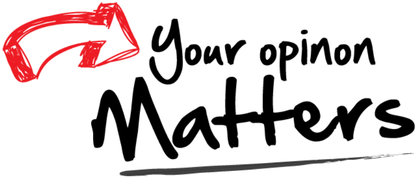
$$\mathrm{Ker}\ \phi = \mathbb{R}$$

$$\mathbb{R}[x]/\mathbb{R} \cong \phi(\mathbb{R}[x]) = \mathbb{R}[x]$$


Thank
You!



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